

From a Gravity-Calibrated Helix to Neutrino Oscillation Correspondences: A Third Cross-Sector Comparison of the Frozen TEF Shape Parameter

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Abstract

This paper is the third cross-sector comparison in a TEF sequence built around the same dimensionless helical shape parameter $q = a/R = 0.556324180045\dots$. The value of q was fixed in Paper I by a gravity-calibrated one-cycle condition, before any neutrino observable entered the analysis, and was inherited without reoptimization by Paper II in a conditional fine-structure correspondence.

Motivated by the electrically neutral and colorless character of active neutrinos, the present work carries the same frozen q into neutrino oscillation data. Two post-hoc but coefficient-free correspondences are recorded: $q^2 = 0.3094965933\dots$, compared with JUNO's $\sin^2 \theta_{12} = 0.3092 \pm 0.0087$, and $q^6 = 0.0296461034\dots$, compared with $r_\nu = \Delta m_{21}^2 / |\Delta m_{3\ell}^2|$. Current JUNO and NuFIT 6.1 values give $r_\nu \simeq 0.02975$ for normal ordering and $r_\nu \simeq 0.02988$ for inverted ordering, both compatible with q^6 at present precision. Eliminating q gives the prospective phenomenological test $r_\nu \simeq \sin^6 \theta_{12}$; this is not an independent third piece of evidence.

No neutrino mechanism is claimed. For the Euclidean circular-helix ansatz, q also satisfies the exact Frenet–Serret identity $q = \tau/\kappa$, giving it an intrinsic dimensionless geometric meaning. A future theory would need a common relativistic spectral operator whose eigenvectors and mass-squared eigenvalue gaps generate the two frozen correspondences without inserting them by hand.

Keywords: neutrino oscillations; helical spacetime; PMNS mixing; Frenet–Serret geometry; flavor symmetry; topology; emergent spacetime; TEF

1 Introduction

The present paper is most naturally read as the third member of a sequence, not as a stand-alone proposal for neutrino physics. The sequence has followed one methodological rule: a geometric quantity should first be fixed within the TEF ansatz and only then exposed to an external sector. The first paper introduced a gravity-calibrated helical spacetime ansatz and obtained a dimensionless shape parameter q before comparing its intrinsic angle with electroweak weak mixing [1]. The second paper inherited exactly the same q , did not reoptimize it, and

asked whether a separate electromagnetic quantity could be organized by the same geometry [2]. That second step was explicitly presented as a conditional fine-structure correspondence rather than a derivation of QED.

The present work applies the same discipline to neutrinos. The question is not whether TEF can already derive the PMNS matrix. It cannot. The narrower question is whether the publicly frozen helical shape produces simple derived numbers that coincide with independently measured neutrino observables closely enough to pose a new dynamical problem.

In the standard three-neutrino framework,

$$|\nu_\alpha\rangle = \sum_{i=1}^3 U_{\alpha i}^* |\nu_i\rangle, \quad \alpha = e, \mu, \tau, \quad (1)$$

where U is the PMNS matrix [3]. Oscillation experiments establish that flavor states are not propagation eigenstates and measure two independent mass-squared splittings together with three mixing angles; the atmospheric neutrino result from Super-Kamiokande was a decisive early demonstration of this phenomenon [4]. JUNO has now entered a precision regime for the solar sector, reporting from its first 59.1 days of data [5]

$$\sin^2 \theta_{12} = 0.3092 \pm 0.0087, \quad (2)$$

$$\Delta m_{21}^2 = (7.50 \pm 0.12) \times 10^{-5} \text{ eV}^2. \quad (3)$$

The NuFIT 6.1 update provides a current global context for the larger splitting and mass ordering [6].

Neutrinos were selected as the next TEF test for a qualitative reason. Active neutrinos carry no electric charge and no color charge; within the Standard Model they are therefore a particularly clean probe of the weak sector. Since the preceding TEF papers had already moved the same q from a gravity-calibrated microscopic shape into electroweak and electromagnetic numerical regimes, the neutrino sector was a natural place to look next. The numerical mappings reported below were nevertheless recognized after this sector was examined and therefore carry a mapping-choice/look-elsewhere burden of their own.

This distinction—a frozen upstream parameter but post-hoc downstream mapping—is central to the interpretation of the paper. The result is neither a fitted neutrino model nor a preregistered prediction. It is a cross-sector numerical observation with fixed upstream ancestry.

2 The TEF sequence and the inherited quantity

2.1 Paper I: gravity calibration and a frozen helical shape

Paper I, *A Helical Spacetime Ansatz Linking the Planck Scale and Electroweak Mixing*, introduced the local circu-

lar helix [1]

$$\mathbf{r}(\phi) = (R \cos \phi, R \sin \phi, a\phi), \quad q \equiv \frac{a}{R} = \tan \beta. \quad (4)$$

Its transverse scale is calibrated by the TEF relation

$$R_G = \sqrt{\frac{hG}{8\pi c^3}} = \frac{\ell_P}{2}. \quad (5)$$

The ansatz then distinguishes the one-turn axial extension from the internal helical path length and imposes a one-cycle phase closure. After inserting the gravity calibration, the dimensional constants cancel from the shape equation, leaving

$$q\sqrt{1+q^2} = \frac{2}{\pi}. \quad (6)$$

The positive solution is

$$q = 0.556324180045\dots \quad (7)$$

and

$$\beta = \arctan q = 29.0882455^\circ\dots, \quad (8)$$

$$\sin^2 \beta = 0.2363477652\dots \quad (9)$$

No electroweak observable is used to obtain Eq. (7). Paper I therefore treated β only as a candidate primitive geometric reference angle for weak mixing; it did not identify a measured, scheme-dependent weak angle directly with β .

2.2 Paper II: a conditional electromagnetic continuation

Paper II, *From a Gravity-Calibrated Helix to a Conditional Fine-Structure Correspondence*, explicitly inherited the same value of q from Paper I and kept it frozen [2]. The exact one-turn helical path excess is

$$\delta_h \equiv \sqrt{1+q^2} - 1 = 0.144332378858\dots \quad (10)$$

That paper then separated exact geometry from additional modeling assumptions: a framed $SU(2) \simeq S^3$ orientation completion, a Hopf-coordinate identification, a unit-round metric/measure convention, and an explicitly named electromagnetic postulate. Under those conditions it proposed

$$\alpha_{\text{TEF}} = \frac{\sqrt{1+q^2} - 1}{2\pi^2}, \quad (11)$$

which yields $\alpha_{\text{TEF}}^{-1} = 136.7621663\dots$, about 0.2002% from the 2022 CODATA low-energy value. Paper II deliberately froze Eq. (11) rather than introduce a correction factor to improve the residual.

This epistemic structure is inherited here. In compact form,

$$G_{\text{exp}} \longrightarrow R_G \longrightarrow q, \quad (12)$$

after which the same frozen shape is carried into the weak-angle candidate of Paper I, the conditional electromagnetic correspondence of Paper II, and the neutrino cross-sector comparison of the present paper. The third branch does not retroactively validate the first two. Its interest is that all three branches reuse one publicly released shape rather than assigning a new continuous fit parameter to each sector.

3 Neutrino observables in the current experimental convention

For relativistic vacuum propagation, the standard oscillation phase contains

$$\Delta\phi_{ij} \simeq \frac{\Delta m_{ij}^2 L}{2E}, \quad \Delta m_{ij}^2 = m_i^2 - m_j^2. \quad (13)$$

The present paper does not require a microscopic interpretation of m_i as a primitive TEF object. It uses the reported mass-squared differences as operationally defined propagation invariants.

To avoid switching definitions between normal and inverted ordering, define

$$r_\nu \equiv \frac{\Delta m_{21}^2}{|\Delta m_{3\ell}^2|}, \quad (14)$$

with the NuFIT convention

$$\Delta m_{3\ell}^2 = \begin{cases} \Delta m_{31}^2, & \text{normal ordering (NO)}, \\ \Delta m_{32}^2, & \text{inverted ordering (IO)}. \end{cases} \quad (15)$$

The first JUNO result fixes the numerator in Eq. (14). For a conservative comparison of the denominator, we use the NuFIT 6.1 global values quoted without the added atmospheric χ^2 tables [6]:

$$\Delta m_{31}^2|_{\text{NO}} = 2.521_{-0.018}^{+0.026} \times 10^{-3} \text{ eV}^2, \quad (16)$$

$$\Delta m_{32}^2|_{\text{IO}} = -2.510_{-0.023}^{+0.024} \times 10^{-3} \text{ eV}^2. \quad (17)$$

This choice is useful because the NuFIT analysis in this

variant uses the published JUNO information for the solar parameters but does not use JUNO's fast-oscillation information to determine $\Delta m_{3\ell}^2$.

The resulting central ratios are

$$r_\nu^{\text{NO}} = 0.0297501, \quad (18)$$

$$r_\nu^{\text{IO}} = 0.0298805. \quad (19)$$

A conservative rectangular endpoint envelope of the quoted one-dimensional marginal errors gives

$$r_\nu^{\text{NO}} = 0.029750_{-0.000775}^{+0.000693}, \quad (20)$$

$$r_\nu^{\text{IO}} = 0.029880_{-0.000745}^{+0.000771}. \quad (21)$$

For IO, taking the absolute value of the negative Δm_{32}^2 reverses the direction of its quoted asymmetric errors. The endpoint envelope is not a standard one-dimensional 68% confidence interval. If the published marginals are instead treated provisionally as independent near-Gaussian inputs, linear quadrature gives the orientation estimates

$$r_\nu^{\text{NO}} \simeq 0.0297501_{-0.000566}^{+0.000521}, \quad (22)$$

$$r_\nu^{\text{IO}} \simeq 0.0298805_{-0.000551}^{+0.000557}. \quad (23)$$

Neither approximation replaces a joint likelihood or covariance analysis.

4 Cross-sector numerical observations

4.1 Observation I: the solar mixing number

The first simple descendant of the frozen shape is

$$q^2 = 0.309496593303\dots \quad (24)$$

which may be placed beside JUNO's Eq. (2). The central-value difference is

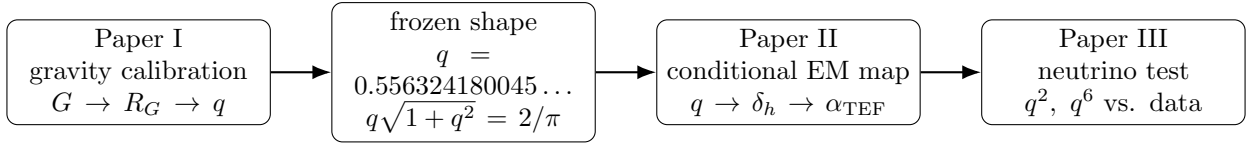
$$q^2 - 0.3092 = 2.97 \times 10^{-4}, \quad (25)$$

about 0.096% of the JUNO central value and approximately 0.034 of its quoted 1σ uncertainty.

The corresponding amplitude-level rewriting is

$$q \simeq \sin \theta_{12}. \quad (26)$$

Equation (26) is not derived from a neutrino Hamiltonian.



The neutrino comparison inherits the same publicly released q ; it does not re-estimate the shape from neutrino data.

Figure 1: Logical continuity of the three TEF papers. Arrows denote inheritance or an explicitly conditional continuation, not a completed derivation of the corresponding Standard-Model sector.

It is the minimal reading of Eq. (24) if q is associated with an amplitude whose experimentally reported quantity is squared. A future theory would have to produce the relevant interaction/propagation basis overlap rather than insert it by definition.

4.2 Observation II: the splitting hierarchy

The sixth power of the same shape is

$$q^6 = 0.029646103401 \dots \quad (27)$$

Compared with Eqs. (18)–(19), the differences in central values are approximately 0.35% for NO and 0.78% for IO. Both are comfortably inside current uncertainties; the present comparison therefore has no useful mass-ordering discrimination.

If one uses the JUNO value of Δm_{21}^2 as a scale input, the correspondence $r_\nu \simeq q^6$ implies for NO

$$\Delta m_{31}^2 \simeq \frac{\Delta m_{21}^2}{q^6} = 2.52984 \times 10^{-3} \text{ eV}^2, \quad (28)$$

close to Eq. (16). Conversely, using the NuFIT NO denominator gives

$$\Delta m_{21}^2 \simeq q^6 \Delta m_{31}^2 = 7.474 \times 10^{-5} \text{ eV}^2, \quad (29)$$

close to Eq. (3). These are conditional restatements of one ratio correspondence, not independent predictions.

4.3 The parameter-eliminated relation

If the two observations share one origin, eliminating q yields

$$r_\nu \simeq \sin^6 \theta_{12}. \quad (30)$$

Equivalently,

$$\sqrt{r_\nu} \simeq \sin^3 \theta_{12}. \quad (31)$$

This is the cleanest phenomenological statement because it can be tested without accepting TEF. It is not a third independent numerical success: it follows algebraically from the two frozen mappings after eliminating q and is retained for its value as a prospective parameter-eliminated test. With the first JUNO central value,

$$(\sin^2 \theta_{12})^3 = 0.02956095. \quad (32)$$

A linear propagation of JUNO’s present θ_{12} uncertainty gives roughly 0.00250, much wider than the uncertainty on the splitting ratio. Thus the current agreement of Eq. (30) is suggestive but statistically weak; the relation becomes more discriminating primarily as θ_{12} precision improves.

4.4 Discovery-order and search-space disclosure

The sequence of inference matters:

1. q was fixed in Paper I and publicly released before the present neutrino comparison;
2. Paper II reused the same q in an electromagnetic correspondence;
3. the neutrino sector was then selected on qualitative electroweak grounds;
4. the mappings q^2 and q^6 were recognized after examining neutrino oscillation observables.

Hence the value of q is out-of-sector, but the mapping functions are not preregistered predictions. This paper does not assign a formal look-elsewhere p -value because the historical search space was not defined prospectively.

It instead freezes the two relations at the present stage:

Table 1: Cross-sector comparison from the frozen TEF shape. Experimental quantities are not used to reoptimize q . The mappings q^2 and q^6 were recognized after the neutrino sector was examined and therefore remain phenomenological.

Quantity	TEF descendant	TEF value	Current comparison
Solar mixing	q^2	0.3094965933	JUNO: $\sin^2 \theta_{12} = 0.3092 \pm 0.0087$
NO splitting ratio	q^6	0.0296461034	$r_\nu^{\text{NO}} = 0.029750^{+0.000693}_{-0.000775}$
IO splitting ratio	q^6	0.0296461034	$r_\nu^{\text{IO}} = 0.029880^{+0.000771}_{-0.000745}$
Conditional NO large splitting	$\Delta m_{21}^2/q^6$	$2.52984 \times 10^{-3} \text{ eV}^2$	NuFIT 6.1: $2.521^{+0.026}_{-0.018} \times 10^{-3} \text{ eV}^2$

Table 2: Frozen mapping ledger from the present paper onward. These choices define the prospective comparison and are not to be retuned after future data updates.

Item	Frozen choice
Upstream parameter	$q = 0.556324180045 \dots$
Angle observable	$\sin^2 \theta_{12}$
Angle map	q^2
Splitting observable	$\Delta m_{21}^2/ \Delta m_{3\ell}^2 $
Splitting map	q^6
Convention	NuFIT 3ℓ convention
Ordering	NO and IO both reported
Future retuning	none

no alternative exponent, multiplicative correction, or new function of q is introduced to repair future disagreement.

5 The geometric status of the frozen parameter

5.1 A stronger intrinsic identity: $q = \tau/\kappa$

For the circular helix in Eq. (4), the Frenet–Serret curvature and torsion are

$$\kappa = \frac{R}{R^2 + a^2}, \quad (33)$$

$$\tau = \frac{a}{R^2 + a^2}. \quad (34)$$

Therefore

$$q = \frac{a}{R} = \frac{\tau}{\kappa}. \quad (35)$$

For the Euclidean circular-helix ansatz in $\mathbb{R}^3$, this identity is exact differential geometry and does not depend on the neutrino comparison. It strengthens the interpretation of q : the frozen TEF number is not only a pitch-to-radius ratio but also the dimensionless Frenet–Serret ratio of

local twisting to local bending. No claim is made here that this Euclidean curve ratio is already a Lorentzian spacetime invariant.

The symbol τ here is *Frenet–Serret torsion of an embedded curve*. It must not be confused with spacetime torsion in Riemann–Cartan or Einstein–Cartan geometry, where torsion is a tensor associated with an affine connection. The two notions share terminology but are mathematically distinct. This distinction is especially important because there is already a neutrino literature on Einstein–Cartan torsion, discussed below.

5.2 Why q^3 is only a clue

The identity

$$q^6 = (q^3)^2 \quad (36)$$

invites a three-dimensional interpretation. An isotropic map $(x, y, z) \mapsto (qx, qy, qz)$ has Jacobian q^3 . However, a volume Jacobian and an eigenvalue-gap amplitude are different mathematical objects. No result in the present TEF ansatz establishes

$$q^3 = \sqrt{\frac{\Delta m_{21}^2}{|\Delta m_{3\ell}^2|}}. \quad (37)$$

Accordingly, the decomposition $q^6 = (q^3)^2$ is retained only as a suggestive three-dimensional geometric clue. In the absence of a derived spectral connection, it is not treated as a theoretical explanation of the neutrino splitting ratio.

5.3 The spectral problem created by the observations

The natural mathematical target is no longer another numerical formula. It is a single operator derived from the same three-dimensional geometry. To avoid ambiguity

about whether an eigenvalue is a mass, a frequency, or already a squared invariant, it is useful to state the future target directly in terms of a self-adjoint effective mass-squared or propagation-invariant operator,

$$\boxed{\mathcal{M}_{\text{TEF}}^2(q)\Psi_i = \mu_i(q)\Psi_i.} \quad (38)$$

A successful continuation would have to produce both the reduced solar-sector mixing overlap and the appropriate splitting hierarchy.

In the standard three-flavor PMNS convention, $|U_{e2}| = s_{12}c_{13}$ rather than s_{12} itself. The basis-invariant target corresponding to the present q^2 observation is therefore

$$\boxed{\frac{|U_{e2}|}{\sqrt{|U_{e1}|^2 + |U_{e2}|^2}} = \sin \theta_{12} = q,} \quad (39)$$

not the unnormalized statement $|U_{e2}| = q$. Any two-state matrix written with entries $\sqrt{1-q^2}$ and q should consequently be interpreted only as a reduced solar-sector rotation, not as the full PMNS matrix.

For normal ordering, the corresponding spectral target is

$$\boxed{\frac{\mu_2 - \mu_1}{\mu_3 - \mu_1} = q^6,} \quad (40)$$

whereas in the NuFIT inverted-ordering convention it becomes

$$\boxed{\frac{\mu_2 - \mu_1}{|\mu_3 - \mu_2|} = q^6.} \quad (41)$$

Equations (39)–(41) are target conditions, not deductions of the present paper.

Quantum mechanics on curved space curves provides a useful mathematical precedent without supplying this neutrino operator. Ortix derived an effective Schrödinger–Pauli equation for a spin-orbit-coupled electron constrained to a space curve and found curvature/torsion-dependent geometric terms, including an explicit helical case [13]. More recently, Hou, Gao, and Guo showed that retaining a degenerate transverse subspace in a curved quantum waveguide can make Frenet-frame rotation generate a matrix-valued gauge connection controlled by curve torsion [14]. These systems are nonrelativistic constrained waveguides, not neutrinos and not evidence for TEF. Their relevance is narrower: they demonstrate that Frenet–Serret data can enter a genuine quantum operator and its spectrum. The existence of torsion-dependent terms in such constrained systems does not select the powers q^2 or q^6 and therefore does not by itself explain either correspondence reported here.

6 External research context

The word “geometric” covers several very different research programs. To locate the present paper accurately, it is useful to separate at least five of them.

6.1 Flavor symmetry and modular symmetry: the mainstream structural benchmark

Most theoretical attempts to explain lepton mixing do not begin from a literal three-dimensional curve. They construct neutrino and charged-lepton mass matrices from flavor symmetries, residual symmetries, seesaw structures, texture constraints, or related mechanisms. Modular flavor symmetry is a particularly active modern example: the modular parameter and modular forms constrain Yukawa structures and mixing matrices, often together with generalized CP symmetry [7].

This literature provides the appropriate benchmark for what a full TEF neutrino theory would eventually need to achieve. A mature model should not merely reproduce one angle or one splitting ratio; it should specify the state space and operator or mass structure from which the PMNS matrix, eigenvalue hierarchy, CP properties, and matter-dependent propagation follow. The present work does not meet that benchmark and does not claim to.

6.2 Geometric phenomenology of oscillation parameters

A closer methodological comparator is Seikh’s 2025 geometric hypothesis for neutrino oscillation parameters [8]. That work proposes a Euclidean triangle relation among the three mixing angles, treats it explicitly as an empirical phenomenological organizing principle rather than a result of known dynamics, and performs a consistency test against global-fit data.

The similarity is the willingness to state a compact, falsifiable geometric relation before a microscopic mechanism is known. The difference is the ancestry of the organizing quantity. Seikh constructs the geometric relation directly from neutrino mixing angles, whereas the present q was defined and numerically fixed in a separate gravity-calibrated helix construction before the neutrino comparison. This does not remove mapping-choice freedom, but it makes the present test genuinely cross-sector.

6.3 Topology and geometric phases in neutrino physics

Topology has also entered neutrino theory in more microscopic ways. Gu proposed a model in which neutrinos are built from topological Majorana zero modes and used the construction to address three generations and mixing [9]. Separately, geometric and topological phases have been studied within otherwise standard neutrino oscillation kinematics, including two-flavor topological phases and three-flavor gauge-invariant geometric phases [10, 11].

These works establish that neither “topology and neutrinos” nor “geometry and oscillation phase” is a novelty claim available to TEF. Their mechanisms are, however, distinct from the present proposal. The TEF observation concerns the reuse of one dimensionless invariant of a prior microscopic helical-spacetime ansatz and does not presently derive a Berry/Pancharatnam phase or a Majorana zero-mode construction.

6.4 Spacetime torsion and the terminological boundary

There is also an active literature on neutrino oscillations in backgrounds with spacetime torsion. For example, Panda *et al.* study modifications of the oscillation Hamiltonian from torsional couplings in curved spacetime and assess long-baseline sensitivities [12]. Such models use torsion/contorsion of the spacetime affine connection, commonly in an Einstein–Cartan-type setting.

Equation (35) concerns something different: τ is the Frenet–Serret torsion of the TEF circular helix. The present work therefore does not cite Einstein–Cartan torsion as a derivation or precedent for $q = \tau/\kappa$; it cites that literature to mark the distinction and to avoid a false equivalence.

6.5 The 2026 Zenodo geometry–PMNS landscape

Zenodo is a general-purpose repository rather than a peer-review filter. Its records are nevertheless useful for novelty checks because speculative geometry and neutrino models often appear there before, or without, journal publication. A targeted 2026 search finds several nearby examples.

McKoy’s *Neutrino Masses and PMNS Mixing Angles from $SO(6)$ Boundary Geometry* uses an $SO(6)$ -based

geometric construction to quote all three PMNS angles and an absolute neutrino mass scale [15]. Jagadeesan’s OOB series likewise reports geometry-driven, nominally parameter-free PMNS values and explicitly compares standing predictions with the first JUNO result [16]. These deposits are useful comparators for the current speculative landscape, not validation benchmarks.

The lesson for the present paper is methodological. Geometry-based numerical claims are easy to proliferate, especially when many algebraic expressions or structural constants are available. The TEF series therefore restricts its narrow claim to the portability of one frozen upstream parameter and states explicitly where a downstream mapping was recognized after the fact.

6.6 Narrow novelty boundary

A targeted, necessarily non-exhaustive search of arXiv and Zenodo through 29 August 2026 did not locate the exact phenomenological relation

$$\frac{\Delta m_{21}^2}{|\Delta m_{3\ell}^2|} \simeq \sin^6 \theta_{12} \quad (42)$$

or the specific use of the gravity-calibrated circular-helix invariant

$$q = \frac{a}{R} = \frac{\tau}{\kappa} \quad (43)$$

as the common antecedent of the weak-angle, fine-structure, and neutrino correspondences developed in the three TEF papers. This is a limited novelty statement about the present search, not a claim of priority for geometric or topological approaches to neutrino mixing in general.

7 Questions raised rather than solved

The strongest version of the present result is a question, not an explanation. The two numerical observations ask whether one microscopic geometric degree of freedom can appear both in a state overlap and in a propagation spectrum.

7.1 Can the helix generate the basis change?

If Eq. (26) has physical content, a theory must explain why the Frenet–Serret ratio τ/κ becomes the overlap

between an interaction-defined electron-neutrino state and a propagation eigenmode. Writing by hand the two-state matrix

$$\begin{pmatrix} \sqrt{1-q^2} & q \\ -q & \sqrt{1-q^2} \end{pmatrix} \quad (44)$$

is algebraically consistent but does not answer the question.

7.2 Can the same operator generate the gap ratio?

The stronger constraints are Eqs. (40) and (41). The same operator must possess three relevant eigenmodes and must produce the measured hierarchy without assigning the exponent six after seeing the data. This requires a boundary-value problem, an energy functional or relativistic wave operator, and a clear identification of the measured oscillation phases.

7.3 Why three active modes?

The three-neutrino framework is an experimental input here, not a TEF result. A true topological completion should first classify the admissible modes and only then identify three of them with ν_1, ν_2, ν_3 . The same classification would need to explain why the remaining PMNS parameters are not arbitrary.

7.4 What would make the spacetime-mode hypothesis physical?

One possible TEF interpretation, motivating but not required by the numerical result, is that a neutrino is a propagating spinorial mode associated with emitted spacetime structure. To become physics rather than ontology, such a proposal must recover at least the observed relativistic dispersion, weak chirality, W/Z couplings, standard oscillation probabilities, matter effects, and the absence of observable preferred-frame effects. These derivations are beyond the scope of the present paper.

8 Methodological boundaries and falsifiability

8.1 Freeze the upstream quantity and the present mappings

The value

$$q = 0.556324180045 \dots \quad (45)$$

is inherited from Paper I and is not to be refitted to neutrino data. From the publication of the present analysis onward, the two reported mappings are also treated as frozen empirical targets:

$$\sin^2 \theta_{12} \stackrel{?}{=} q^2, \quad r_\nu \stackrel{?}{=} q^6. \quad (46)$$

Future disagreement should not be repaired by changing the exponent, choosing a new trigonometric function, or adding a fitted coefficient.

8.2 Do not confuse cross-sector ancestry with a preregistered prediction

Because q was fixed upstream, the comparison is stronger than defining q from neutrino data. But the functions q^2 and q^6 were recognized downstream. Accordingly, the paper uses phrases such as “cross-sector correspondence” and “coefficient-free conditional relation” rather than claiming a parameter-free first-principles prediction.

8.3 Use ordering-safe and convention-explicit comparisons

The ratio r_ν in Eq. (14) is retained precisely to prevent a silent switch among Δm_{31}^2 , Δm_{32}^2 , $\Delta m_{3\ell}^2$, and Δm_{ee}^2 . Both NO and IO are shown. Any future precision update should use public likelihoods/covariances where available rather than treating all reported errors as independent Gaussians.

8.4 No search for the remaining PMNS parameters in this paper

No new expression in q is proposed for θ_{13} , θ_{23} , or δ_{CP} . Nor is q used here to fit the absolute neutrino mass scale. The next meaningful advance is an operator or topological classification that outputs additional observables without a new post-hoc search.

9 Discussion

The three-paper TEF sequence now has a clearer logical shape. Paper I fixed a microscopic helix using a gravity calibration and a cycle condition. Paper II asked whether that frozen shape admits a conditional electromagnetic numerical map, while explicitly separating exact geometry from postulates and conventions. The present paper does not add a new primitive constant. It asks whether the same shape leaves a further trace in neutrino oscillation data.

The answer, as a numerical statement, is compact:

$$q^2 = 0.3094965933 \dots \quad (47)$$

lies near the measured $\sin^2 \theta_{12}$, while

$$q^6 = 0.0296461034 \dots \quad (48)$$

lies near the measured small-to-large oscillation splitting ratio. If both are physical manifestations of the same structure, they imply

$$r_\nu \simeq \sin^6 \theta_{12}. \quad (49)$$

At current precision this relation is compatible with the data but weakly tested, primarily because the uncertainty propagated from θ_{12} remains broad.

An additional geometric observation is not another fitted formula but the exact identity $q = \tau/\kappa$ for the Euclidean circular-helix ansatz. It converts the inherited shape parameter into a ratio of the two Frenet–Serret invariants of the circular helix. Existing curved-curve quantum mechanics shows that curvature and torsion can enter actual quantum operators. The research problem is therefore sharply stated: determine whether a relativistic, spinorial, covariant TEF operator built from the same geometry can produce both the required eigenvector overlap and the required eigenvalue-gap ratio. Until then, the two successful numbers remain phenomenology.

This placement also clarifies the relationship to the surrounding literature. Mainstream flavor and modular-symmetry programs provide the standard of a true mixing model; geometric phenomenology shows that compact empirical relations can be scientifically testable before a mechanism is known; topology and geometric phases are already established themes in neutrino research; and torsion-based neutrino studies use a different mathematical torsion from the one entering the TEF helix. Zenodo contains additional 2026 geometry–PMNS proposals,

which reinforces the need for narrow novelty claims and explicit mapping-choice accounting.

10 Conclusion

A gravity-calibrated helical-spacetime ansatz fixed the dimensionless shape

$$q = 0.556324180045 \dots \quad (50)$$

in the first TEF paper, before neutrino data were considered in this research sequence. A second TEF paper inherited the same shape in a conditional fine-structure correspondence. The present third paper carries that frozen quantity into neutrino oscillation phenomenology and records

$$q^2 = 0.3094965933 \dots \simeq \sin^2 \theta_{12}, \quad (51)$$

$$q^6 = 0.0296461034 \dots \simeq \frac{\Delta m_{21}^2}{|\Delta m_{3\ell}^2|}. \quad (52)$$

The parameter-eliminated statement is

$$\boxed{\frac{\Delta m_{21}^2}{|\Delta m_{3\ell}^2|} \simeq \sin^6 \theta_{12}}. \quad (53)$$

It can be tested independently of TEF as neutrino precision improves.

The paper does not claim that neutrino mixing has been explained. Its geometric addition is the exact identity

$$q = \frac{a}{R} = \frac{\tau}{\kappa} \quad (54)$$

for the inherited circular helix, which turns the frozen number into a Frenet–Serret invariant ratio and motivates, but does not supply, a spectral mechanism. The unresolved problem is whether one operator derived from that same three-dimensional structure can generate both the relevant state overlap and the propagation-gap hierarchy.

The appropriate status of the result is therefore deliberately modest: a cross-sector numerical correspondence produced by a previously released geometric parameter, embedded in the contemporary neutrino literature and converted into a specific dynamical question. If future data reject either frozen relation, the correspondence fails. If a common spectral mechanism can be derived without reoptimizing the mappings, the numerical observation would acquire a qualitatively different theoretical status.

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